envguard

Zero-Boilerplate Environment Variable Validation

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Malformed Env Vars Threaten Stability

Modern Node.js applications frequently fail due to misconfigured environment variables. These silent issues, ranging from runtime crashes to security vulnerabilities, interrupt production reliability and create significant debugging overhead for development teams.

Deferred Runtime Crashes

Applications boot successfully, only to crash hours later when internal logic finally attempts to access an undefined or missing environment variable.

Type Mismatch Errors

Because process.env returns strings, logic often fails, such as the string "false" evaluating as truthy or arithmetic operations failing with numeric strings.

Critical Security Risks

Sensitive keys like JWT_SECRET may be accidentally set to short or weak strings without developer detection, leaving the application vulnerable to exploits.

Fragmented Validation

Traditional validation stops at the first error found, forcing developers into a tedious, slow cycle of fix-restart-fail loops during each deployment attempt.

EnvGuard Configuration Safety

EnvGuard solves configuration fragility by moving validation to the very first line of your application, ensuring your environment is correct before execution begins.

Startup Validation

If your environment is wrong, the app will not start, preventing any unstable states in production.

Automatic Coercion

Seamlessly transforms string inputs into true JavaScript types like numbers, booleans, and strings.

Security Enforcement

Uses minLength checks to ensure all environment secrets meet the required entropy requirements.

EnvGuard Setup

The envguard implementation uses a zero-boilerplate approach, requiring only a single function call at the entry point of your script to secure your environment variables effectively and cleanly.

Import & Initialize

Call the guard() function at the top of your index.js file right after loading dotenv.

Schema Definition

Pass an object where keys represent variables and values define rules like type and default.

Validation & Mapping

The library iterates through your schema, checks process.env, and performs type conversions.

Unified Error Handling

EnvGuard throws a single, formatted error listing every violation instead of failing once.

Technical Architecture & Specifications

A lightweight footprint built for the modern ESM ecosystem

Pure JavaScript (ESM)

Utilizes pure ECMAScript modules for native import syntax support and optimized runtime performance.

Zero Dependencies

No external packages beyond optional dotenv, reducing vulnerability and security surface area.

MIT License

Open source and fully unrestricted for use in both commercial enterprise and public software projects.

NPM Package

Available through the npm registry as @rohansm14/envguard for simple standard project installation.

Envguard's Unique Advantages Over Traditional Validators

01

Actionable Error Output

Envguard replaces cryptic stack traces with clear, visual reports, detailing specific missing variables, length requirements, and incorrect data types like strings versus numbers.

02

Zero-Boilerplate Design

It eliminates the need for complex classes or decorators. Developers manage all validation logic using a straightforward, easy-to-read, JSON-like schema structure.

03

Smart Defaults

Easily assign fallback values for non-required variables, ensuring your application runs smoothly in development environments without needing extensive manual configuration.

Enterprise Scale

EnvGuard Roadmap

Advanced Types

- Validate JSON strings
- Email address parsing
- Regex pattern support

Phase 1

Custom Validators

- Inject custom logic
- Domain-specific checks
- Custom validation flows

Phase 2

DevOps Tooling

- Auto-generate .env.example
- Sync docs with schemas
- Eliminate drift

Phase 3

Performance

- Negligible startup load
- Handle 50+ variables
- Efficient validation

Phase 4

Start guarding your Node.js environment today

Stop runtime crashes and start building with confidence using envguard.